

Mental Health Signal

Machine Learning Information Report

Purpose

This project estimates an informational Mental Health Score from student social-media habits, academic routine, lifestyle signals, and perceived stress. The output is a reflection tool, not a clinical assessment or diagnosis.

Dataset

The source file contains 4,998 records and 13 input features plus the target Mental_Health_Score. Ages range from 18 to 24; the average target score is 6.23/10. The target ranges from 3.6 to 9.4. Duplicate rows were removed and negative physical-activity values were clipped to zero.

Features and preprocessing

Study_Hours is log-transformed with log1p and scaled. Other numeric variables are standardized. Stress_Level is encoded with the ordered categories Low, Medium, High, and Very High. Gender, academic level, platform, purpose, and grouped country are one-hot encoded with unknown categories ignored. Countries outside the top ten are grouped as Other.

Evaluation

Model	Test R ²	Train R ²	MAE	RMSE
Linear Regression	0.743	0.723	0.532	0.670
Random Forest	0.876	0.981	0.348	0.465
Random Forest tuned	0.864	0.955	0.369	0.488

Model used by the app

The API loads Mental_Health_Model.pkl, a RandomForestRegressor pipeline with the preprocessing described above. The notebook also evaluates a tuned random forest using randomized search with 15 configurations and five-fold cross-validation. Metrics describe this dataset split and should not be treated as clinical validity evidence.

RAG use

Signal Guide retrieves relevant passages from this report and the wellness knowledge base before asking Mistral to answer. Mistral is instructed to stay within retrieved context, avoid diagnosis, and recommend qualified human support for urgent concerns.

Limitations and safety

The model reflects correlations in this dataset and can be wrong for an individual. It does not assess safety, symptoms, treatment, or emergencies. For urgent risk, contact local emergency services or a crisis line and reach a trusted person or qualified professional.